

Gurimaal — Real Estate & Utility Management Module

Architecture & ERD Documentation Hierarchy, with End-to-End System Workflow

Frappe Framework Custom App | Includes the new TENANT entity (User-linked, auto-creates Customer)

This document describes the module architecture, the entity-relationship (ERD) hierarchy, and the end-to-end operational workflow of the Gurimaal real estate and utility billing system, built on the Frappe Framework. It reflects the latest model update, which introduces a dedicated **TENANT** DocType sitting between the portal **User** account and the accounting **Customer** record, so that every signed Contract is tied to a tenant identity rather than directly to a Customer.

1. System Overview

The system manages the full lifecycle of rental real estate: property hierarchy, tenant onboarding, lease contracts, utility metering and billing, sales and accounting documents, and maintenance operations. It is organized into five functional domains that map directly onto Frappe DocType modules:

Domain	Purpose	Core DocTypes
Property Structure	Physical hierarchy of real estate assets, from project down to individual unit.	Real Estate Project, Rental Building, Rental Floor, Rental Unit
Tenancy & CRM	Identity and legal relationship between people and the property they occupy or pay for.	Tenant, User, Customer, Contract, Auto Repeat
Utility Infrastructure	Metering, tariff structures, and billing configuration for utility services.	Utility Property, Utility Bill Structure, Bill Structure Item, Billing Adjustment Rule, Serial No, Meter Reading, Utility Billing Settings
Service & Finance	Commercial documents that turn occupancy and consumption into revenue.	Utility Service Request, Sales Order, Sales Invoice, Payment Entry
Maintenance	Reactive and preventive upkeep of units and shared infrastructure.	Maintenance Request, Maintenance Job, Maintenance Checklist, Vendor

What changed: previously, Contract.tenant linked straight to Customer. The new TENANT DocType becomes the identity record a lease is signed against. TENANT links to a Frappe User (for portal login) and, on creation, automatically generates the linked Customer used for all invoicing and payment behind the scenes — Contract, Utility Service Request, and Maintenance Request now read through TENANT rather than requiring staff to manage Customer records directly.

2. ERD Documentation Hierarchy

The diagram below expresses the same five domains as a parent/child documentation tree — the structure to follow when writing Codex prompts or onboarding a new developer, moving from the physical asset down to the financial and operational documents it produces.

1. Property Structure (top-down physical hierarchy)

- Real Estate Project → *has* → Rental Building
- Rental Building → *has* → Rental Floor
- Rental Floor → *has* → Rental Unit
- Rental Unit is the leaf node referenced by Contract, Utility Service Request, Meter Reading, and Maintenance Request

2. Tenancy & CRM (identity → legal agreement)

- User (Frappe core) → *linked by* → Tenant
- Tenant → *auto-creates* → Customer (on insert, if Utility Billing Settings.auto_create_customer is checked)
- Tenant → *signs* → Contract (rental_property, rental_unit, monthly_rent, escalation rules)
- Contract → *configured on* → Auto Repeat (drives recurring rent invoice generation)
- Contract → *generates* → Sales Order (deposit order, from security_deposit_amount)

3. Utility Infrastructure (metering & tariff configuration)

- Utility Property → *located at* → Rental Unit
- Serial No (meter) → *meters at* → Meter Reading
- Meter Reading → *uses tariff* → Utility Bill Structure → *blocks* → Bill Structure Item
- Billing Adjustment Rule → *adjusts* → Meter Reading totals
- Utility Billing Settings holds company-wide defaults (accounts, cost center, auto_create_customer, require_contract)

4. Service & Finance (occupancy/consumption → revenue)

- Customer / Tenant → *submits* → Utility Service Request → *orders* → Sales Order
- Meter Reading → *triggers* → Sales Order / Sales Invoice
- Sales Order → *invoiced* → Sales Invoice → *paid via* → Payment Entry
- Contract recurring rent → Auto Repeat → *spawns* → Sales Invoice (periodic)

5. Maintenance (issue → resolution)

- Customer/Tenant → *raises* → Maintenance Request (unit, property, category, priority)
- Maintenance Request → *spawns* → Maintenance Job → *assigned to* → Vendor
- Maintenance Job → optional link → Purchase Order (cost reconciliation)
- Rental Unit → *inspected via* → Maintenance Checklist (periodic inspection log)

3. Entity Reference

Key fields and purpose for the DocTypes most relevant to the tenant on-boarding change. Full field lists for every DocType are kept in the live ERD (Section 6).

3.1 TENANT (new)

Field	Type	Notes
name	Data (PK)	Naming series, e.g. TEN-.YYYY.-.#####
tenant_name	Data	Full legal name of the tenant
user	Link → User	Unique; the portal login account for this tenant
customer	Link → Customer	Read-only; auto-populated on insert
mobile_no / email	Data	Email can fetch from user.email
national_id	Data	KYC reference
customer_group	Link	Used when generating the Customer; falls back to Utility Billing Settings default
status	Select	Active / Inactive / Blacklisted

3.2 Auto-create Customer logic (Tenant controller hook)

- Triggered on **after_insert** of Tenant.
- Skip creation if a Customer is already linked (idempotent — prevents duplicates on re-save or import).
- New Customer: customer_name = tenant_name, customer_type = Individual, customer_group = tenant.customer_group or Utility Billing Settings default if blank.
- The linked User is added to the Customer's Portal Users child table, so the tenant can log in and see their own Sales Invoices and Payment Entries.
- tenant.customer is set to the new Customer's name and saved.
- Respects Utility Billing Settings.auto_create_customer — if unchecked, Tenant is saved without a Customer and staff must link one manually before a Contract can be invoiced.

4. End-to-End System Workflow

The operational flow below traces a single unit from setup through occupancy, billing, and maintenance, showing where the Tenant entity now sits in the chain.

PHASE 1	Property Setup
1	Admin creates Real Estate Project, then Rental Building(s) under it (total_floors recorded).
2	Rental Floor records are created per building (floor_number).
3	Rental Unit records are created per floor — unit_type, area_sqft, bedrooms, monthly_rent, status = Vacant.
PHASE 2	Tenant Onboarding (updated flow)
1	Front-desk or self-service portal creates a Frappe User account for the prospective tenant.
2	Staff creates a Tenant record and links the User; tenant_name, mobile_no, national_id, customer_group are captured.
3	On save, the system auto-creates the linked Customer (per Section 3.2) — no manual Customer entry needed.
4	The Tenant's User is now able to log into the portal and will see invoices/payments once a Contract exists.
PHASE 3	Contract & Leasing
1	Staff creates a Contract: rental_property, rental_unit, tenant (Link → Tenant), monthly_rent, security_deposit_amount, escalation_percentage / escalation_interval, notice_period_days.
2	Contract generates a deposit Sales Order for security_deposit_amount, invoiced and paid before move-in.
3	Contract is configured on an Auto Repeat record (frequency = Monthly, start_date = contract_start_date) to spawn recurring rent Sales Invoices automatically.
4	next_escalation_date drives scheduled rent increases at the configured escalation_interval.
5	Rental Unit.status is updated to Occupied once the Contract is active.
PHASE 4	Utility Service & Metering
1	Tenant/Customer submits a Utility Service Request for a unit and service_type (e.g. electricity, water).
2	The request orders a Sales Order (connection/activation charge, if applicable) and links the Contract.
3	A meter (Serial No) is installed and recorded against the Utility Property; recurring Meter Readings are logged.
4	Each Meter Reading uses the assigned Utility Bill Structure; consumption is priced through Bill Structure Item slabs (from_units/to_units, rate_per_unit, fixed_charge).
5	Billing Adjustment Rules apply any standing adjustments (discounts, surcharges) by priority before the Meter Reading triggers Sales Invoice generation.
PHASE 5	Invoicing & Payment
1	Sales Orders (deposit, rent, service, utility) are invoiced into Sales Invoice records.
2	Utility Billing Settings supplies default income account, receivable account, cost center, and invoice status.

3	Tenant/Customer pays via Payment Entry — mode_of_payment and reference_name reconcile against the invoice.
4	Portal-linked Tenant Users can view their own Sales Invoice and Payment Entry history at any time.

PHASE 6	Maintenance
1	Tenant/Customer raises a Maintenance Request against their unit (category, priority, description).
2	The request spawns a Maintenance Job, assigned to a Vendor, with estimated_cost vs actual_cost tracked.
3	Completed jobs can link to a Purchase Order for vendor cost reconciliation.
4	Rental Units are periodically inspected via Maintenance Checklist records (inspection_type, remarks).

5. Key Business Rules & Configuration

Setting / Rule	Where	Effect
auto_create_customer	Utility Billing Settings	Controls whether saving a Tenant automatically creates its linked Customer.
require_contract	Utility Billing Settings	If checked, a Utility Service Request cannot be processed without an active Contract on the unit.
Idempotent Tenant→Customer	Tenant controller	Re-saving a Tenant never creates a duplicate Customer; existing link is reused.
Escalation schedule	Contract	escalation_percentage applied to monthly_rent at each escalation_interval, updating next_escalation_date.
Tariff slabs	Bill Structure Item	Consumption priced progressively between from_units and to_units at rate_per_unit, plus fixed_charge.
Recurring rent	Auto Repeat	frequency / start_date / end_date determine when the next rent Sales Invoice is spawned from the Contract.

6. Appendix — Full ERD Diagram

Reference snapshot of the complete entity-relationship diagram, including the new Tenant entity (top half shown here; continued on the next page).

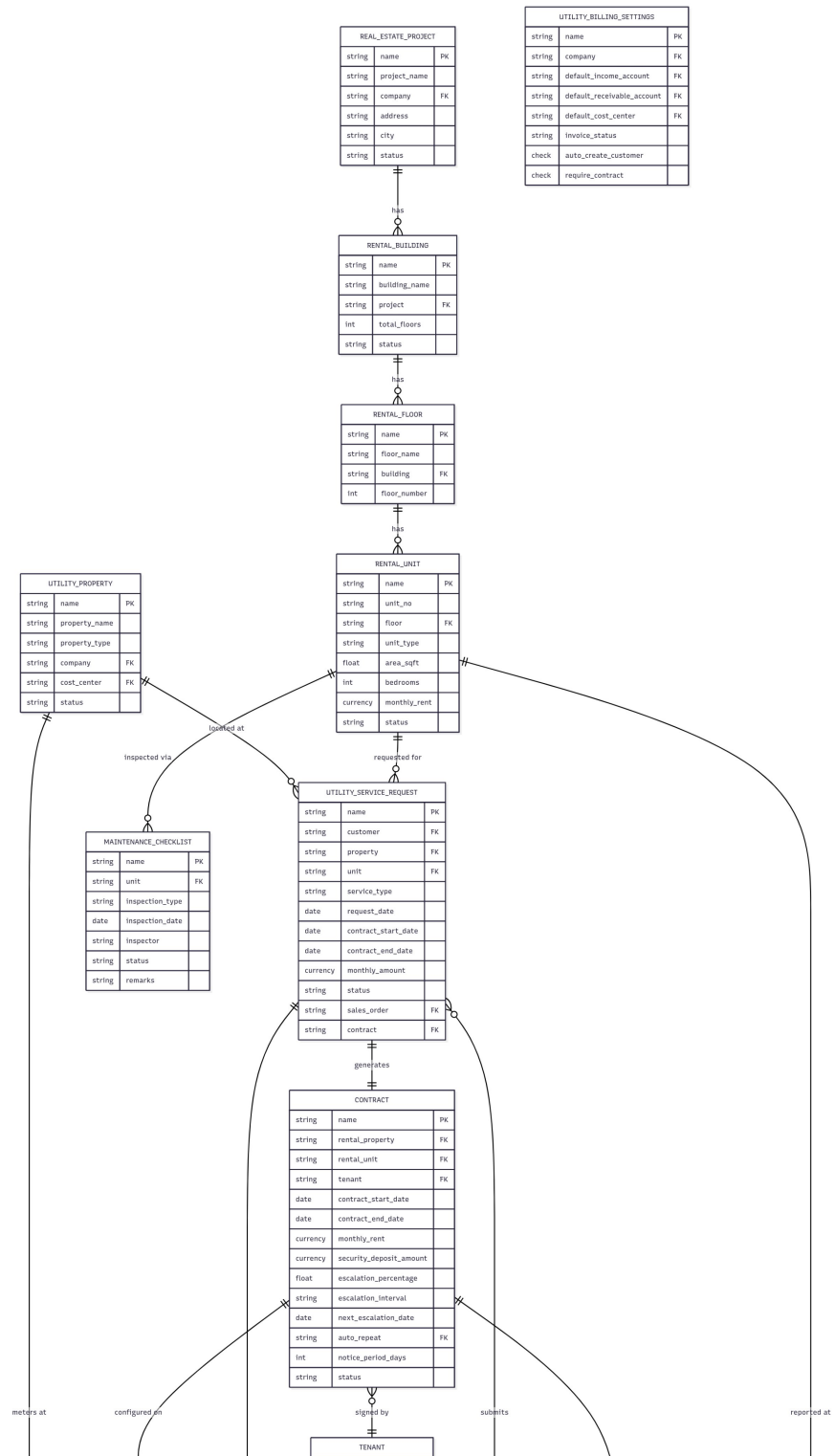


Figure 1 — Property structure, utility infrastructure, and Tenant / Contract relationships.

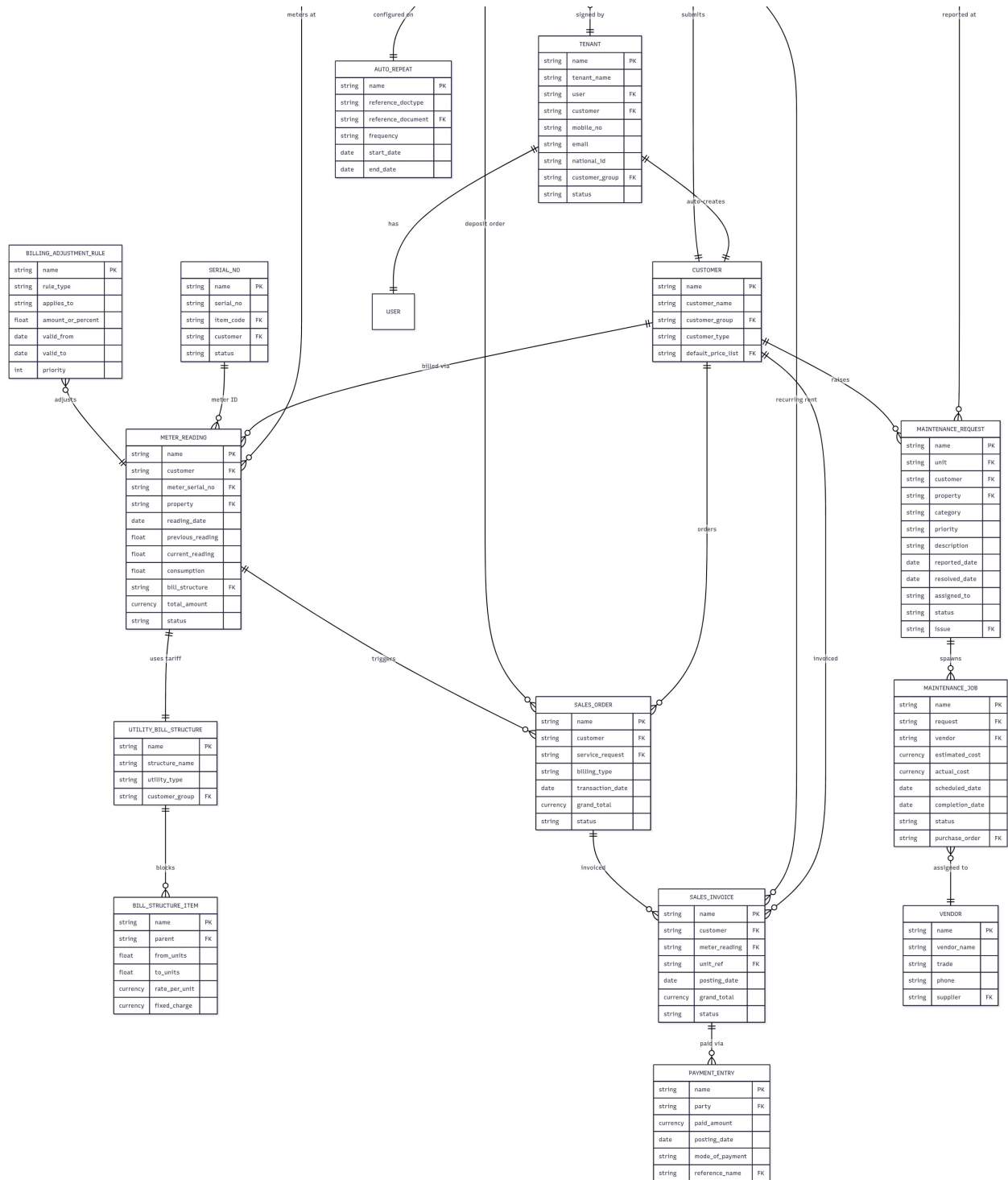


Figure 2 — Service, billing, payment, and maintenance relationships.